VANISHING OF MAPS FROM A PERFECT OBJECT IN SMALL DEGREE IS UNAFFECTED BY UPPER TRUNCATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The canonical truncation triangle tau_{<= a} E -> E -> tau_{> a} E -> gives a long exact sequence on Hom groups. used at proof lines 10706-10712. Relates the two Hom-vanishing statements.
- Maps from a perfect object into an object of D^b_QCoh vanish in small degree after shifting. invoked at proof lines 10712-10713. Gives the needed vanishing for the upper truncation.

Source reference. Original lemma 09M4: sources/stacks-perfect/pinned-original-source.tex, source lines 3465–3465. Cited at proof lines 10712-10713 for the vanishing into the upper truncation after shifting; the reduction via the truncation triangle is inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $P \in D_{perf}(\mathcal{O}_X)$ and $E \in D_{QCoh}(\mathcal{O}_X)$. Let $a \in \mathbf{Z}$. The following are equivalent*

(1) $\mathrm{Hom}_{D(\mathcal{O}_X)}(P[-i], E) = 0$ *for* $i \ll 0$, *and*
(2) $\mathrm{Hom}_{D(\mathcal{O}_X)}(P[-i], \tau_{\leq a}E) = 0$ *for* $i \ll 0$.

Proof. Using the triangle $\tau_{\leq a}E \to E \to \tau_{>a}E \to$ we see that the equivalence follows if we can show

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(P[-i], \tau_{>a}E) = \mathrm{Hom}_{D(\mathcal{O}_X)}(P, (\tau_{>a}E)[i]) = 0$$

for $i \ll 0$. As P is perfect this is true by Lemma 09M4. □

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